The Gate Challenge

A Predeclared Falsification Protocol for MAAT v1.7 Gate-Aware Structural Selection

MAAT Structural Selection Series – Paper 68

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Abstract

MAAT v1.7 introduced gate-aware structural selection: the distinction between score coordinates and gate coordinates. The central hypothesis is that some structural diagnostics, especially emergent robustness, may be more useful as conditional activation variables than as continuously weighted score terms. This remains a framework hypothesis, not an established principle.

This paper defines a predeclared falsification protocol for MAAT v1.7. The default assumption is that the gate hypothesis is false until it improves practical decision utility on external benchmarks. The protocol freezes the gate equation, calibration rule, domain polarity, metrics, baselines, null controls, bootstrap procedure, and success criteria before external experiments are run.

The primary test is not whether MAAT correlates with a diagnostic target. The primary test is whether the same gate architecture improves decisions relative to both score-based MAAT and strong classical baselines across independent domains. The first three primary domains are JHTDB-style turbulence refinement, public SAT/CDCL benchmarks, and externally specified open quantum-system benchmarks.

The protocol is deliberately adversarial. MAAT v1.7 receives only minimum support if the gate beats score-based MAAT in at least two of three primary external domains with paired-bootstrap confidence intervals excluding zero. A stronger v2.0-style claim is forbidden unless the gate also beats the strongest classical baselines in all three primary domains. Failure is an admissible outcome. The strong-support criterion is intentionally severe and should be read as a long-term validation target, not as the expected outcome of the next single benchmark paper.

1 Purpose

Paper 68 is not a new MAAT version. It is a challenge protocol. Its purpose is to move MAAT v1.7 from internal pattern recognition to external decision-level testing.

The hypothesis under test is:

> *A predeclared gate-aware structural rule improves practical decisions better than a continuously weighted structural score in multiple independent external domains.*

The default assumption is the opposite:

> *MAAT v1.7 is false until supported by predeclared external utility tests.*

This is important because the earlier v1.7 paper was a framework refinement motivated by converging internal benchmark evidence. That evidence was useful, but insufficient. The present paper fixes the next test before the data are used.

The criteria below are intentionally asymmetric. Minimum support asks whether the gate improves over the fair score-based comparison. Strong support asks whether the gate also defeats the best classical domain-native baselines. The latter is deliberately hard and should be treated as a long-term standard for any future v2.0-style claim.

2 What Is Being Tested

MAAT v1.7 distinguishes a score from a gate:

$$\text{score coordinate} \neq \text{gate coordinate}.$$

A score is graded. A gate is conditional.

Earlier score-based MAAT forms can be written schematically as

$$F[X] = F(H, B, S, V, R_{\text{rob}}; X), \tag{1}$$

where all structural diagnostics enter one global ranking or utility functional. The gate-aware alternative is

$$F_{\text{active}}[X] = G_{\text{gate}}[X]\, F_{\text{score}}[H, B, S, V; X], \tag{2}$$

where G_{gate} decides whether the structural channel is active.

The test is therefore not:

Does a MAAT score correlate with the target?

The test is:

Does a predeclared gate improve decisions?

3 Frozen Gate Architecture

The following gate architecture is frozen for the Gate Challenge. Let X denote a candidate state, search node, time window, refinement cell, or diagnostic sample. The primary structural supports are H, B, S, V, and the emergent robustness closure is

$$R_{\text{resp}}[X] = (H[X]B[X]V[X])^{1/3}, \qquad R_{\text{rob}}[X] = \min\Big(R_{\text{resp}}[X], (H[X]B[X]S[X]V[X])^{1/4}\Big). \tag{3}$$

For each domain D, a polarity $p_D \in \{+1, -1\}$ is predeclared:

$$p_D = +1 \quad \text{means high robustness activates a preservation/selection mode,}$$

$$p_D = -1 \quad \text{means low robustness activates a warning/refinement mode.}$$

The calibration fold defines

$$m_D = \text{median}_{\text{cal}}(R_{\text{rob}}), \qquad s_D = \text{MAD}_{\text{cal}}(R_{\text{rob}}) + \epsilon, \qquad \epsilon = 10^{-9}.$$

The normalized robustness gate coordinate is

$$z_R[X] = p_D\, \frac{R_{\text{rob}}[X] - m_D}{s_D}. \tag{4}$$

The response-aware gate includes temporal and spatial response terms when available:

$$g[X] = z_R[X] + a_t z_t[X] + a_x z_x[X], \qquad a_t = 0.50, \quad a_x = 0.25. \tag{5}$$

Here

$$z_t[X] = p_D\, \partial_t \widehat{R_{\mathrm{rob}}}[X], \qquad z_x[X] = p_D\, \nabla \widehat{R_{\mathrm{rob}}}[X],$$

with hats denoting standardization on the calibration fold only. If a domain has no meaningful temporal or spatial response term, the corresponding term is set to zero before evaluation.

The soft gate is

$$\boxed{G_{\mathrm{gate}}[X] = \frac{1}{1+\exp[-g[X]/\tau]}, \qquad \tau = 1.} \tag{6}$$

The hard gate used for discrete decisions is

$$\boxed{G_{\mathrm{gate}}^{\mathrm{hard}}[X] = \mathbf{1}[G_{\mathrm{gate}}[X] \geq 0.5].} \tag{7}$$

No test-set retuning of m_D, s_D, a_t, a_x, τ, polarity, or the hard-gate threshold is allowed.

4 Predeclared Domains

The primary evidence must come from external or independently specified benchmarks. Internal MAAT artifacts may be used only for debugging the implementation.

Domain	**Polarity**	**Decision**	**Primary metric**
Fluid / JHTDB turbulence	-1	Trigger adaptive refinement or early warning	Lead utility at matched false-alarm rate
SAT / CDCL	$+1$	Activate structural branching or fall back to classical heuristic	Paired compute regret
Open quantum systems	$+1$	Select robust quantum states	Ranking utility or top-k enrichment

Table 1: Primary Gate Challenge domains. The gate architecture is identical; only the task polarity is predeclared.

An optional secondary domain is public machine-learning sample hardness (sklearn/OpenML-style datasets). It can be used as an additional stress test, but it cannot replace one of the three primary domains.

These domains are deliberately heterogeneous. That heterogeneity is useful because it makes accidental domain-specific success less persuasive, but it also makes interpretation harder. A two-of-three result should therefore be interpreted as evidence for conditional portability of the gate architecture, not as proof of a universal law.

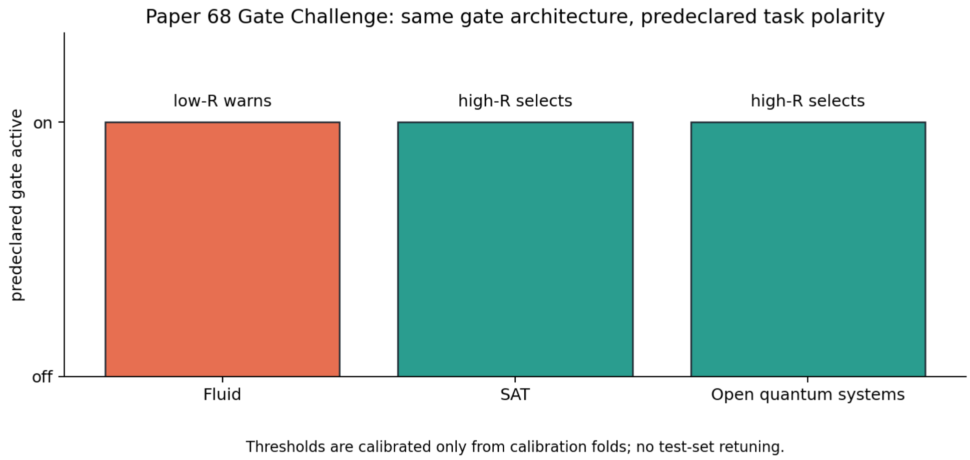

Figure 1: Predeclared Gate Challenge structure. The same response-aware gate is used across domains; only the decision polarity is fixed in advance by the task type.

5 Baselines

Every domain must compare three model classes:

1. the strongest reasonable classical baseline;
2. score-based MAAT;
3. gate-aware MAAT.

Domain	**Required baselines**
Fluid	high-k monitor, palinstrophy, max vorticity, RMS vorticity
SAT	VSIDS, LRB, MOMS, Jeroslow–Wang
Quantum	decoherence rate, purity, fidelity, stationarity baseline
ML hardness	local label disagreement, kNN density, centroid distance, PCA error

Table 2: Required baseline families. A gate result is not meaningful unless it is compared against strong domain-native monitors or heuristics.

If a stronger baseline is available for a specific public benchmark, it should be added before evaluation. Baselines may not be weakened to make the gate look better.

6 Evaluation Protocol

6.1 Pre-registration

Before running a benchmark, the following must be fixed:

- gate equation;
- polarity;
- calibration split;
- threshold rule;

- primary and secondary metrics;
- classical baselines;
- null controls;
- bootstrap procedure;
- failure criteria.

6.2 Gate versus Score

For each domain, evaluate:

$$U_{\text{classical}}, \qquad U_{\text{score}}, \qquad U_{\text{score}+R}, \qquad U_{\text{gate}},$$

where U is the predeclared domain utility. The primary comparison is

$$\Delta U_{\text{gate}-\text{score}} = U_{\text{gate}} - U_{\text{score}+R}. \tag{8}$$

This is the fair internal comparison. The gate is not primarily being compared against a score that lacks robustness. It is being compared against the best score-based use of the same robustness information. The stronger comparison is

$$\Delta U_{\text{gate}-\text{best}} = U_{\text{gate}} - \max_i U_{\text{classical},i}. \tag{9}$$

6.3 Bootstrap and Null Controls

All reported utility deltas must include paired bootstrap 95% confidence intervals. The minimum number of bootstrap resamples is $N_{\text{boot}} = 1000$.

The following null controls are mandatory where applicable:

- shuffled R_{rob} within domain;
- shuffled gate signal preserving its marginal distribution;
- random polarity assignment;
- target-label shuffle for labelled prediction tasks.

6.4 Ablation

Every benchmark must include:

- value-only gate $g = z_R$;
- response-only gate $g = a_t z_t + a_x z_x$;
- full response-aware gate, Equation (5);
- hard gate versus soft gate;
- score-only versus score-with-R versus gate-with-R.

7 Success and Failure Criteria

The protocol distinguishes minimum support, strong support, and failure.

Outcome	**Criterion**
Minimum support	Gate beats score-with-R in at least two of three primary external domains with paired-bootstrap 95% lower bound > 0, and no primary domain has an upper bound < 0.
Strong support	Gate beats both score-with-R and the strongest classical baseline in all three primary domains with paired-bootstrap 95% lower bound > 0.
Failure	Gate fails to beat score-with-R in at least two primary domains, or improvements vanish under null controls, or gains require post-hoc threshold tuning.
v2.0 claim	Forbidden unless strong support is met on external benchmarks.

Table 3: Predeclared interpretation of Gate Challenge outcomes.

Negative results must be reported. If the gate loses, the correct conclusion is not to invent a new gate after the fact. The correct conclusion is that MAAT v1.7 remains unsupported in that domain or under that decision type.

8 Why This Is Paper 68

The previous v1.7 note established the conceptual distinction between scores and gates. Paper 68 turns that distinction into a falsification protocol.

This is a necessary step because internal benchmark convergence can be misleading. SAT, quantum, fluid, and transfer benchmarks suggested a gate-aware pattern, but those observations were not enough to establish a principle. The Gate Challenge asks whether the pattern survives:

- external data,
- strong baselines,
- no post-hoc tuning,
- identical gate architecture,
- explicit failure criteria.

9 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle

Protocol folder:

`experiments/maat_v17_gate_challenge_paper68/`

Run:

```
cd experiments/maat_v17_gate_challenge_paper68
python3 gate_challenge_protocol.py
```

Generated protocol artifacts:

- `gate_challenge_preregistration.json`;
- `gate_challenge_domain_matrix.csv`;
- `gate_challenge_baseline_matrix.csv`;
- `gate_challenge_metric_registry.csv`;
- `fig1_gate_challenge_protocol.png`.

Data and license note. Paper 68 introduces no external raw dataset and reports no external benchmark result. It defines a preregistered protocol for future external validation. Future JHTDB use must follow current JHTDB access terms and cite the relevant JHTDB publications. Future SAT, quantum, OpenML, or other public datasets must retain their original attribution and license requirements. No endorsement by any external dataset provider is implied.

10 Conclusion

The Gate Challenge is designed to make MAAT v1.7 vulnerable. That is its purpose.

The framework receives no credit for internal correlations unless they improve external decisions under a predeclared gate architecture. The same gate must be tested against score-based MAAT and strong classical baselines. Failures must be reported.

If the gate wins across external domains, MAAT v1.7 gains real support. If it does not, the gate hypothesis remains an interesting but unsupported framework refinement. No v2.0 claim should be made unless the stronger success criterion is met.

References

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